Major phase-involved modulation schemes

Memoryless and **Inherent Memory** Schemes

- **Memoryless schemes** (like standard PSK or QAM) map a block of bits directly to a single symbol; the receiver makes independent, symbol-by-symbol decisions. They do not require trellis decoding natively (unless an external error-correction code like Trellis Coded Modulation is added).
- **Schemes with inherent memory** (like Continuous Phase Modulation) restrict how the signal transitions from one symbol to the next to keep the phase continuous. Because the current phase state depends on previous bits, the optimal receiver hardware must look at a sequence of symbols over time using Maximum Likelihood Sequence Estimation (MLSE), which is implemented via a Trellis diagram and the Viterbi algorithm.

Phase Modulation Classification and Demodulation Matrix

Modulation Scheme	Detection Method	Trellis-Based Decoding (Native)	Core Concept
M-PSK (BPSK, QPSK, 8-PSK)	Coherent	**No** (Symbol-by-symbol)	Maps discrete bit sequences to specific, absolute phase angles on a carrier wave while holding amplitude constant.
OQPSK & $\pi/4$-QPSK	Coherent	**No** (Symbol-by-symbol)	Variations of QPSK that offset timing or rotate the constellation to prevent the signal envelope from dropping to zero during phase transitions.
DPSK & DQPSK	Non-Coherent	**No** (Symbol-by-symbol)	Encodes data in the phase *difference* between the current symbol and the immediately preceding symbol, eliminating the need for absolute phase tracking.
M-QAM (16-QAM, 64-QAM)	Coherent	**No** (Symbol-by-symbol)	A hybrid scheme that simultaneously modulates the independent amplitudes of an in-phase (I) and a quadrature (Q) carrier, resulting in points varying in both amplitude and absolute phase.
M-APSK (16-APSK, 32-APSK)	Coherent	**No** (Symbol-by-symbol)	Arranges constellation points in concentric rings with varying phase and

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			amplitude; highly resilient to the nonlinear distortions found in satellite amplifiers.
CPFSK (Continuous Phase FSK)	Coherent or Non-Coherent	**Yes** (Optimal)	Encodes information as frequency shifts while strictly enforcing a continuous phase boundary between symbols. The phase trajectory contains memory of past bits.
MSK (Minimum Shift Keying)	Coherent or Non-Coherent	**Yes** (Optimal)	A specific type of CPFSK where the phase changes by exactly $\pm 90^\circ$ ($\pm\pi/2$ radians) over one bit period, providing continuous phase and compact spectrum.
GMSK (Gaussian MSK)	Coherent or Non-Coherent	**Yes** (Optimal)	Passes the data through a Gaussian low-pass filter before MSK modulation, further smoothing the phase transitions and heavily restricting out-of-band spectral leakage.

A Note on Hardware Implementation:

While M-PSK and M-QAM are listed as "No" for native trellis decoding, modern communication hardware rarely transmits these schemes without forward error correction. In practice, high-order QAM and PSK are frequently combined with Trellis Coded Modulation (TCM) at the transmitter. When TCM is applied, the receiver hardware *will* utilize a trellis decoder (Viterbi decoder) to unravel both the modulation and the error-correction coding simultaneously. However, the trellis decoding in that scenario is a function of the applied channel coding, not an inherent requirement of the modulation physics itself.